SRINIVAS UNIVERSITY

INSTITUTE OF COMPUTER SCIENCE & INFORMATION SCIENCES

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Question Bank Answers with Bloom's Model

HTML, VBSCRIPT AND JAVASCRIPT
COURSE CODE: 25BCA32

B.C.A - III SEMESTER

Compiled by

Faculty

MODULE-I

Multiple Choice Questions(1 Marks):-

1. What is the full form of HTML?

   A. Hyper Text Markup Language
   B. High Text Markup Language
   C. Hyper Tabular Markup Language
   D. None of these

2. Which of the following tag is used to mark the beginning of a paragraph?

   A. <TD>
   B.
   C. <P>
   D. <TR>

3. Which HTML tag is used to insert an image into a web document?

   A. <imp>
   B. <img>
   C. <src>
   D. <images>

4. Which file extension is used to save an HTML file?

   A. htnl
   B. htl
   C. hml
   D. .html

5. Which of the following tag is used to create a numbered list?

   A. <LI>
   B. <OL>
   C. <UL>
   D. None of these

6. Which attribute specifies the image path in the <img> tag?

   A. Href
   B. Src
   C. Alt
   D. Link
   Answer: B. Src

7. What is the purpose of the <title> tag?
   A. To create a heading
   B. To display the page title in the browser tab
   C. To add an image
   D. To create a hyperlink

Answer: B. To display the page title in the browser tab

8. Who is considered the inventor of HTML?
A) Bill Gates
B) Dennis Ritchie
C) Tim Berners-Lee
D) Brendan Eich

Answer: C. Tim Berners-Lee

9. Which tag is used to create a list item?
   A) <item>
   B)
   C) <list>
   D) <dt>

   Answer: B.<li>

10. Which type of pathname contains the complete web address?
   A) Relative pathname
   B) Nested pathname
   C) Local pathname
   D) Absolute pathname

Answer: D. Absolute pathname

11. **Write an HTML program to insert a line break within a paragraph of text.**

a) Hello
World

b) Hello<hr>World

c) Hello<break>World

d) Hello World

Answer: a) Hello
World

12 **Write an HTML program to create spacing between table cells.**

a) <table cellpadding="10">

b) <table cellspacing="10">

c) <table tablespace="10">

d) <table spacing="10">

Answer: b) <table cellspacing="10">

13. Write an HTML program to change the color of visited links to red.

a) <body link="red">

b) <body alink="red">

c) **<body vlink="red">**

d) <body visited="red">

14. **Write an HTML program to display an image using the <img> tag. Which of the following is correct?**

a) `<img src="image.jpg">`
b) `<img href="image.jpg">`
c) `<image src="image.jpg">`
d) `<img link="image.jpg">`

Answer: a) **`<img src="image.jpg">`**

15 Write an HTML program to display preformatted text.

a) <text>Welcome to HTML</text>

b) <format>Welcome to HTML</format>

c) <pre>Welcome to HTML</pre>

d) Welcome to HTML

Answer: c) <pre>Welcome to HTML</pre>

16 Write the HTML code used to merge two rows in a table cell.

a) <td colspan="2">Data</td>

b) <td rowspan="2">Data</td>

c) <td merge="2">Data</td>

d) <td rows="2">Data</td>

Answer: b) <td rowspan="2">Data</td>

17 Write the HTML code used to open a hyperlink in a new browser tab.

a) Click Here

b) Click Here

c) Click Here

d) Click Here

Answer: a) Click Here

18 Write an HTML <img> tag using an absolute path.

a) <img src="photo.jpg" alt="Photo">

b) <img src="/images/photo.jpg" alt="Photo">

c) <img src="https://example.com/images/photo.jpg" alt="Photo">

d) <img path="photo.jpg" alt="Photo">

Answer: c) <img src="https://example.com/images/photo.jpg" alt="Photo">

19 Write the HTML code to create a table with one row and two columns.

a) <table><tr><td>A</td><td>B</td></tr></table>

b) <table><td><tr>A</tr><tr>B</tr></td></table>

c) <table><row><column>A</column><column>B</column></row></table>

d) <table><tr><td>A</td></tr><tr><td>B</td></tr></table>

Answer: a) <table><tr><td>A</td><td>B</td></tr></table>

20 Write the HTML code to display an image with the alternative text "College".

a) <img src="college.jpg" text="College">

b) <img src="college.jpg" title="College">

c) <img src="college.jpg" alt="College">

d) <image src="college.jpg" alt="College">

Answer: c) <img src="college.jpg" alt="College">

Short Answer Questions(3 Marks) :

1. What are the <html> and <head> tags in HTML?
Answer:

- The <html> tag is the root element of an HTML document. It wraps all the content on the web page, including both visible and invisible elements. All other HTML tags must be written inside the <html> tag.
- The <head> tag contains metadata about the HTML document. This includes the title, links to CSS files, JavaScript, and other information that is not displayed on the page.

2. What is an attribute in HTML?
Answer:

An Attribute in HTML provides additional information about an element. Attributes are always written inside the opening tag**.**

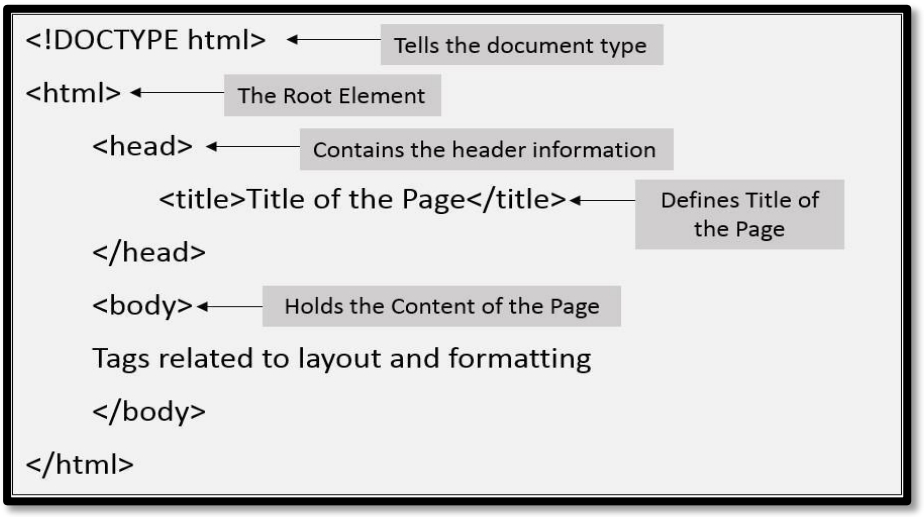
HTML attributes consist of two parts:

- a name and
- a value

Some common examples of HTML attributes are id, class, align, href, src, height, width etc.

3. Write the basic structure of an HTML document.

Answer:



4. What is the HTML tag used to work with images? Explain with an example.

Answer:

In HTML, images are added using the <img> tag. The src attribute specifies the path or URL of the image

file. The alt attribute provides alternative text, which is shown if the image cannot be displayed. You can also set the size of the image using width and height attributes.

Example:

<img src="flower.jpg" alt="A beautiful flower" width="300" height="200">

5. What is a hyperlink in HTML? Give an example.

Answer:

The <A> tag defines a hyperlink, which is used to link from one page to another. The most important attribute of the <A> element is the href attribute, which indicates the link's destination.

By default, links will appear as follows in all browsers:

An unvisited link is underlined and blue
A visited link is underlined and purple
An active link is underlined and red
Ex: Visit W3Schools.com!

6. What is used to merge rows and columns in an HTML table? Give an example.

Answer:

In HTML, to merge table cells, use the attributes **colspan and rowspan.**

- To merge table columns, use the colspan attribute.
- To merge table rows, use the rowspan attribute.

Example:

```
<table border="1">
  <tr>
    <td colspan="2">Merged Columns</td>
  </tr>
  <tr>
    <td rowspan="2">Merged Rows</td>
    <td>Cell 2</td>
  </tr>
  <tr>
    <td>Cell 3</td>
  </tr>
</table>
```

7. What are Absolute URL and Relative URL? Explain with suitable examples.

Answer:

An **absolute** URL contains the full web address including the protocol,domain name, and the full path to the file.

Example: https://www.example.com/images/photo.jpg

A **relative** URL gives the path relative to the current page's location. It does not include the domain name or protocol. It is used to link files within the same website.

Example: images/photo.jpg

8. What is the purpose of the class attribute in HTML?

The **class** attribute is used to **group HTML elements** so that the **same CSS style or JavaScript** can be applied to multiple elements.

Example:

```
<style>
.text {
color: blue;
}
</style>
<p class="text">Hello</p>
<h2 class="text">Welcome</h2>
```

Both the paragraph and heading are displayed in **blue**

9. What is the purpose of the <body> tag?

The <body> tag defines the main content of an HTML webpage. Everything placed inside the <body> tag is displayed in the web browser.

It can contain:

- Headings
- Paragraphs
- Images
- Links
- Tables

Example:

```
<body>
  <h1>Welcome</h1>
  <p>This is my webpage.</p>
</body>
```

10. What are the parts of an HTML table?

An HTML table is used to organize data into rows and columns. The main parts of an HTML table are:

<table> – Defines the entire table.

<tr> – Defines a table row.

<th> – Defines a table header cell (displayed in bold by default).

<td> – Defines a table data cell.

<caption> – Adds a title to the table.

Long Answer Questions (5 Marks):

1. What is HTML? What are the Features of HTML?

Answer

HTML is an acronym which stands for **Hyper Text Markup Language**. Hyper Text simply means "Text within Text". A markup language is a programming language that is used make text more interactive and dynamic.

FEATURES:

- It is a very easy and simple language. It can be easily understood and modified.
- It is very easy to make effective presentation with HTML because it has a lot of formatting tags.
- It facilitates programmers to add link on the web pages (by html anchor tag), so it enhances the interest of browsing of the user.
- It is platform-independent because it can be displayed on any platform like Windows, Linux and etc.
- It facilitates the programmer to add Graphics, Videos, and Sound to the web pages which makes it more attractive and interactive.
- It is a markup language so it provides a flexible way to design web pages along with the text.

2. What is the <body> tag in HTML? List its attributes.

Answer

- The <body> tag is used to display the main content of a web page.
- It contains text, images, links, tables, lists, etc.
- The <body> tag is closed using </body> before the </html> tag.

Common Attributes of <body> tag:

- **bgcolor** – Sets the background color of the web page. Example: <body bgcolor="lightblue">
- **text** – Sets the color of the text. Example: <body text="black">
- **link** – Sets the color of normal (unvisited) links. Example: <body link="blue">
- **vlink** – Sets the color of visited links. Example: <body vlink="purple">
- **alink** – Sets the color of active (clicked) links. Example: <body alink="red">

3. **What are the types of lists in HTML? Give an example.**

Answer:

In HTML, **lists** are used to display items in a structured format. all lists must contain one or more list elements .

There are three types of lists:

1. Unordered List

An unordered list displays items using bullets. We use the <ul> tag to create the list and the <li> tag for each item.

Example:

```
<ul>
  <li>Apple</li>
  <li>Mango</li>
</ul>
```

2. Ordered List

An ordered list displays items using numbers or letters. We use the <ol> tag to create the list and the <li> tag for each item.

Example:

```
<ol>
  <li>Ready</li>
  <li>Set</li>
</ol>
```

3. Definition List

A definition list displays a term and its description. We use three tags:

- <dl> – Creates the definition list.
- <dt> – Specifies the term.
- <dd> – Gives the description of the term.

Example:

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
</dl>
```

4. What is an HTML table? Give an example and mention five attributes of the <table> tag.

Answer:

An HTML table is used to display data in rows and columns.

- <table> – Creates a table.
- <tr> – Creates a table row.
- <th> – Creates a heading cell.
- <td> – Creates a data cell.

Example:

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>John</td>
    <td>25</td>
  </tr>
</table>
```

Five attributes of the <table> tag:

1. border: Sets the table border.
2. cellpadding: Sets the space between content and cell border.
3. cellspacing: Sets the space between cells.
4. width: Sets the width of the table.
5. align: Aligns the table to the left, center, or right.

5. What are the <pre> and <hr> tags in HTML? Give examples.

Answer:

- The <pre> tag defines preformatted text.
- Text in a <pre> element is displayed in a fixed-width font, preserves all spaces, tabs and line breaks.
- It is used to display text exactly as it is written in the HTML file.
- The <hr> tag in HTML is used to create a horizontal line across the webpage
- The <hr> tag is an empty (self-closing) tag, which means it doesn't have a closing tag
- Eg:

```
<html>
<head><title>Pre tag</title>
</head>
<body>
  <pre>
    This    is    preformatted    text.
    It preserves spaces and    line breaks exactly.
  </pre>
    <hr>
    <p>This is the second section separated by a horizontal line.</p>
    </body>
    </html>
```

6. List any five HTML attributes and state their purpose.

HTML attributes provide **additional information** about HTML elements. They are written inside the opening tag.

Attribute	Purpose
id	Uniquely identifies an HTML element.
class	Groups multiple elements to apply the same CSS style or JavaScript.
href	Specifies the destination URL of a hyperlink (<a> tag).
src	Specifies the path or location of an image (<img> tag).
alt	Provides alternative text if the image cannot be displayed.

Example:

```
<a href="https://www.google.com">Google</a>
<img src="flower.jpg" alt="Flower Image">
<p id="para1" class="text">Welcome to HTML</p>
```

7. List the basic HTML tags and state their purpose.

Attribute	Purpose
<html>	Defines the root of an HTML document.
<head>	Contains information about the webpage, such as the title and metadata.
<title>	Specifies the title displayed on the browser tab.
<body>	Contains the content displayed on the webpage.
<h1> to <h6>	Defines headings of different sizes.
<p>	Defines a paragraph.
<a>	Creates a hyperlink.
<img>	Inserts an image into the webpage.
/	Creates an unordered or ordered list.
	Defines a list item.

8. **What is the <img> tag in HTML? Mention its attributes with an example.**

The **<img>** tag is used to insert or display an image on a web page. It is an **empty tag**, which means it does not have a closing tag (</img>).

Syntax:

```
<img src="image.jpg" alt="Sample Image" width="300" height="200">
```

Attributes of the <img> Tag:

1. **src (Source)**
   Specifies the location (path) of the image.
   Example:<img src="flower.jpg">

2. **width**
   Specifies the width of the image.
   Example:<img src="flower.jpg" width="300">

3. **alt (Alternative Text)**
   Displays text if the image cannot be loaded.
   It also improves accessibility.
   Example:<img src="flower.jpg" alt="Flower">

4. **height**
   Specifies the height of the image. Example:<img src="flower.jpg" height="200">

9. **What is an ordered list in HTML? Write its syntax and one example.**

- An **ordered list** in HTML is used to display a list of items in a **specific sequence**. Each item in the list is automatically numbered or lettered. The ordered list is created using the <ol> tag, and each item in the list is enclosed within the <li> (list item) tag.

Syntax:

```
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
  </ol>
```

Example:

```
<html>
<head>
  <title>Ordered List</title>
</head>
<body>
<h2>Programming Languages</h2>
<ol>
  <li>C</li>
  <li>Java</li>
  <li>JavaScript</li>
</ol>
</body>
</html>
```

10. What is the basic structure of an HTML document? Write its syntax and an example.

The **basic structure of an HTML document** consists of the essential tags required to create a webpage.

Structure of an HTML Document

Syntax:

```
<!DOCTYPE html>
<html>
<head>
   <title>Page Title</title>
</head>
<body>
   Content of the webpage</body></html>
```

Example:

```
<!DOCTYPE html>
<html>
<head>
   <title>My First Page</title>
</head>
<body>
   <h1>Welcome to HTML</h1>
   <p>This is my first webpage.</p>
</body>
</html>
```

Description of the Tags

Tag	Purpose
`<!DOCTYPE html>`	Declares that the document is an HTML5 document.
`<html>`	The root element that contains all the HTML content.
`<head>`	Contains metadata such as the title and links to external files.
`<title>`	Specifies the title displayed on the browser tab.
`<body>`	Contains the visible content of the webpage, such as headings, paragraphs, images, and tables.

MODULE-II

Multiple Choice Questions(1 Marks):-

1. Which HTML tag is used to create a drop-down list?

A. <select>
B. <text>
C. <textarea>
D. <dropdown>

2. Which of the following HTML tag is used for creating a multi-line input area?

A. <textarea>
B. <text>
C. <blockarea>
D. <textfields>

3. What is the default value of the type attribute in the <input> element in HTML?

A. Text
B. Password
C. Numerals
D. Special Characters

4. What is the purpose of HTML forms?

A. To display the content of an email.
B. To display animation effect.
C. To collect user input.
D. None of the above

5. Which HTML tag is used for grouping form controls?

A. <select>
B. <legend>
C. <label>
D. <fieldset>

6. What is the opacity value for a completely transparent element?

A) 1
B) 0.5
C) 0
D) 2

7. Which of the following HTML tags is used to create a label for a form element?

A) <input>
B) <label>
C) <form>
D) <legend>

8. Which of the following is a non-semantic HTML tag?

A) <article>
B) <footer>
C) <nav>
**D) **

9. Which CSS property is used to fix an element at a specific position on the screen even when the page is scrolled?

A. position: absolute
B. position: fixed
C. position: relative
D. position: static

10. Which CSS property is used to change the text color of an element?
A. background-color
B. font-style
C. color
D. text-align

11. Which CSS property should be used when you want to increase the size of the text on a webpage?
A) font-style
B) font-size
C) font-weight
D) text-align
Answer: B) font-size

12. Which method should be used to send data more securely?
A) GET
B) LINK
C) POST
D) SEND
Answer: C) POST

13. Which CSS property is used to add space at the top of an element?
A) margin-top
B) padding-left
C) padding-top
D) border-top
Answer: C) padding-top

14. Which HTML tag is used to create a clickable link to another webpage?
A) <p>
B) <a>
C) <link>
D) <nav>
Answer: B) <a>

15. Which method is used by default when an HTML form is submitted?
A) POST
B) PUT
C) GET
D) SEND
Answer: C) GET

16. Which CSS property is used to place an element on the left or right side of its container?
A) position
B) float
C) clear
D) display
Answer: B) float

17. Which CSS property is used to prevent an element from being affected by floating elements?

A) float
B) position
C) clear
D) display
Answer: C) clear

18. Which CSS property is used to make an element partially transparent?

A) visibility
B) opacity
C) display
D) filter
Answer: B) opacity

19. Which HTML tag is used to show the completion progress of a task?

A) <meter>
B) <progress>
C) <status>
D) <range>
Answer: B) <progress>

20. Which CSS property is used to control how an element is positioned on a webpage?

A) float
B) display
C) position
D) align
Answer: C) position

Short Answer Questions(3 Marks) :

1. **Explain absolute positioning in CSS with an example.**
   Answer:

Absolute positioning in CSS removes an element from the normal document flow and positions it relative to the nearest positioned ancestor (with relative, absolute or fixed). It is placed using top, right, bottom, or left properties and does not affect or get affected by sibling elements.

Example:

```
<div style="position: absolute; top: 10px; right: 10px;">
Absolutely Positioned Div
</div>
```

2. **What are text input controls? Explain their types.**
   Answer:
   Text input controls are form elements used to collect user input in the form of text.
   There are three main types

- **Single-line text input**
  Created using <input type="text">used for names, search boxes, etc.
- **Password input**
  Created using <input type="password">hides typed characters for security.
- **Multi-line text input**
  Created using <textarea> used for longer text like comments or messages.
  Example: <textarea rows="4" cols="30"></textarea>

3. **Explain buttons in HTML forms with an example.**

Answer:

Buttons are used in forms to submit data, reset the form or perform custom actions using JavaScript. Buttons are created using the <button> or <input> tag with specific type attributes.

Types of buttons:

Submit button – Sends form data to the server.

<input type="submit" value="Submit">

Reset button – Clears all input fields in the form.

<input type="reset" value="Reset">

Button – Performs custom actions using JavaScript.

<button type="button" onclick="alert('Clicked')">Click Me</button>

4. **Explain the purpose of the <pre> tag in HTML with an example.**

Answer:

The <pre> tag defines preformatted text

Text in a <pre> element is displayed in a fixed-width font, and the text preserves both spaces and line breaks. The text will be displayed exactly as written in the HTML source code.

Eg:

```
<html>
<body>
<h1> The pre-element</h1>
<pre> Text in a pre-element
is displayed in a fixed-width
font, and it preserves
both spaces and line breaks
</pre>
</body>
</html>
```

5. What is the <form> tag in HTML? Explain the different form controls.
Answer:

The **<form> tag** in HTML is used to create a form for collecting user input. It acts as a container for form elements like text fields, radio buttons, checkboxes, and buttons. Data entered in a form can be submitted to a server for processing using the action and method attributes.

- <input type="text"> – for single-line text input
- <input type="password"> – for password fields
- <textarea> = for multi-line text input
- <input type="radio"> – for selecting one option

6. Explain horizontal rules (<hr>) in HTML and show an example with custom CSS.
Answer:
The <hr> tag in HTML is used to insert a horizontal line, typically to separate content. By default, it renders as a simple horizontal line, but you can customize its style using CSS.
Custom CSS Properties for <hr> include:
width – sets the line length
height – sets the line thickness
color or background-color – sets the line color
border – defines the line's border style
margin – adds spacing around the line
Example:
<hr style="width: 50%; height: 4px; background-color: red; border: none;">

7. Explain fixed positioning in CSS with an example.
Answer:
Fixed positioning in CSS means the element is positioned relative to the browser window, not the document. The element remains in the same position even when the user scrolls the page. It is placed using properties like top, right, bottom, or left.
Example:

```
<style>
.fixed-box {
  position: fixed;
  top: 0;
  right: 0;
  width: 200px;
  border: 2px solid #ee3e64;
  padding: 10px;
}
</style>
<div class="fixed-box">Fixed Element</div>
```

8. What do you mean by Placeholder attribute in HTML? Explain with a suitable example.
The **placeholder** attribute in HTML is used to display a **short hint or sample text** inside an input field or textarea. The hint helps users understand what information they should enter. The placeholder text disappears when the user starts typing.

Example:
```
<html>
<head></head><body>
<form>

 <label for="name">Name:</label>
      <input type="text" name="name" placeholder="Enter your name">
</form>
</body>
</html>
```

9. **Explain the <label> tag in HTML. Illustrate with a suitable example.**

The <label> tag in HTML is used to provide a label or description for a form element such as a text box, radio button, or checkbox. It helps users understand what information should be entered and improves accessibility

```
<html><body>

<form>

  <label for="name">Name:</label>

  <input type="text" name="name">

</form>

</body>

</html>
```

10. **Explain character-level formatting elements in HTML with suitable examples.**

Character-level formatting elements in HTML are used to format or style specific words or characters within a paragraph.

Some commonly used character-level elements are:

- <b> – Displays text in **bold**.
- <i> – Displays text in *italic*.
- – Displays **important text**.
- – Displays *emphasized text*.

```
<html>
<body>
<p>This is <b>bold</b> text.</p>
<p>This is <i>italic</i> text.</p>
<p>This is <strong>important</strong> text.</p>
</body>
</html>
```

Long Answer Questions(5 Marks) :

1. **Differentiate between checkbox and radio buttons.**

Answer:

Checkbox	**Radio Button**
Used to select **one or more** options.	Used to select **only one** option from a group.
Created using <input type="checkbox">.	Created using <input type="radio">.
Multiple checkboxes can be selected at the same time.	Only one radio button can be selected at a time.
Each checkbox works independently.	Radio buttons with the same **name** are grouped together.
Example: Hobbies (Reading, Music, Sports).	**Example:** Gender (Male, Female, Other).

2. **Explain the advantages of CSS.**

Answer:

- **Saves Time** – Write CSS once and use it on multiple HTML pages.
- **Faster Loading** – Less HTML code makes web pages load faster.
- **Easy Maintenance** – A single CSS change updates all related web pages.
- **Better Design** – CSS provides more styling options than HTML.
- **Works on Multiple Devices** – Web pages can be displayed properly on computers, mobiles, and tablets.
- **Follows Web Standards** – CSS is the recommended way to style web pages instead of using HTML formatting tags.

3. **Explain the different ways to include CSS style sheets in an HTML page.**

Answer:

CSS can be added to HTML in the following ways:

§ **Inline** - using the style attribute in HTML elements

Example:

```
<html>
<head>
<title>    </title>
</head>
< body>
< h2 style="background-color:red;"> This is a heading</h2>
<p style=”background-color:green”>This is a paragraph</p>
</body>
</html>
```

§ **Internal** - using the <style>element in the <head> section

```
<html><head>
<style>
P{
color:blue;
}
</style>
</head>
<body>
I am Jahnavi. This is Internal Style sheet
<p>Cascading Style Sheets</p>
<p>Cascading Style </p>
</body>
</html>
```

§ **Externa**l - using an external CSS file
Example:
/* This is an external style sheet file. Save this file as mystyle.css*/
h2
{
color: red;
text-align: left; font-size: 8pt;
}

4. Explain the select box control in HTML with an example.
Answer:
Select box Control:
Drop Down Box is used when we have many options available to be selected but only one or two will be selected.
Following is the list of important attributes of <select>:
- name: This is the name for the control.
- size: This can be used to present a scrolling listbox.
- multiple: If set to "multiple" then allows a user to select multiple items from the menu.

Following is the list of important attributes of <option>:
- value: The value that is sent to the server if this option is selected.
- selected: Specifies that this option should be the initially selected value when the page loads.
- label: An alternative way of labelling options.

Example:
```
<html>
<body>
<form action="/cgi-bin/dropdown.cgi" method="post">
<select name="dropdown">
<option value="Maths" selected>Maths</option>
<option value=”Physics”>Physics</option>
<option value=”Chemistry”>Chemistry</option>
</select>
<input type=”submit” value=”Submit”/>
</form>
</body>
</html>
```

5. Explain universal and grouping selectors in CSS with examples.
Answer:
The CSS Universal Selector
The universal selector (*) selects all HTML elements on the page.
Eg:
```
<html><head>
<style>
   *{
   text-align:center;
   color: blue;
   }
   </style></head><body>
   <h1>Hello world</h1>
   <p> Every element on the page will be affected by the style.</p>
   <h2> here also</h2>
   </body></html>
```

The CSS Grouping Selector The grouping selector selects all the HTML elements with the same style definitions. Look at the following CSS code (the h1, h2, and p elements have the same style definitions) h1 { text-align: center; color: red;
}
h2 {
text-align: center;
color: blue;
}
p {
text-align: center;
color: black;
}
It will be better to group the selectors, to minimize the code. To group selectors, separate each selector with a comma.

```
h1, h2, p {
text-align: center;
color: red;
}
```

6. Explain Id and class selectors in CSS with examples.

1. **ID Selector (#)**

α. The **ID selector** is used to style **one unique HTML element**.
β. An ID should be **unique** on a webpage.
χ. In CSS, the ID selector starts with the # symbol.

Syntax:

```
#idname
{
Property:value;
}
```

Example:

```
<html><head><title>ID Selector</title>
<style>
#heading
{
color:blue;
text-align: center;
}
</style></head>
<body>
<h1 id="heading">Welcome to CSS</h1>
</body>
</html>
```

2. **Class Selector (.)**

δ. The **Class selector** is used to style **multiple HTML elements**.
ε. The same class can be used for many elements.
ϕ. In CSS, the class selector starts with the **.** (dot) symbol.

Syntax:

```
.classname {
  property: value;
}
```

Example:

```
<html><head>
<title>Class Selector</title>
<style>
.redtext
{ color:red;
font-size: 20px;
}
</style></head>
<body>
<p class="redtext">This is the first paragraph.</p>
<p class="redtext">This is the second paragraph.</p>

</body>
</html>
```

7. Explain File Select Boxes in HTML with a suitable example.

File Select Box in HTML is created using the <input type="file"> element. It allows users to **browse and select one or more files** from their computer and upload them to a web server through a form.

Explanation

γ. **type="file"**: Creates a file selection box.
η. **name**: Specifies the name of the file input field.
ι. **accept="image/*"**: Allows only image files (such as JPG, PNG, GIF) to be selected.
φ. **accept=".pdf,.doc,.docx"**: Allows only PDF and Microsoft Word documents.
κ. **multiple**: Allows the user to select more than one file.
λ. **type="submit"**: Creates a button to submit the selected files to the server.

```
<html><body>

<h2>File Upload Form</h2>
<form action="" method="post">

Upload Image:
<input type="file" name="image" accept="image/*"><br><br>

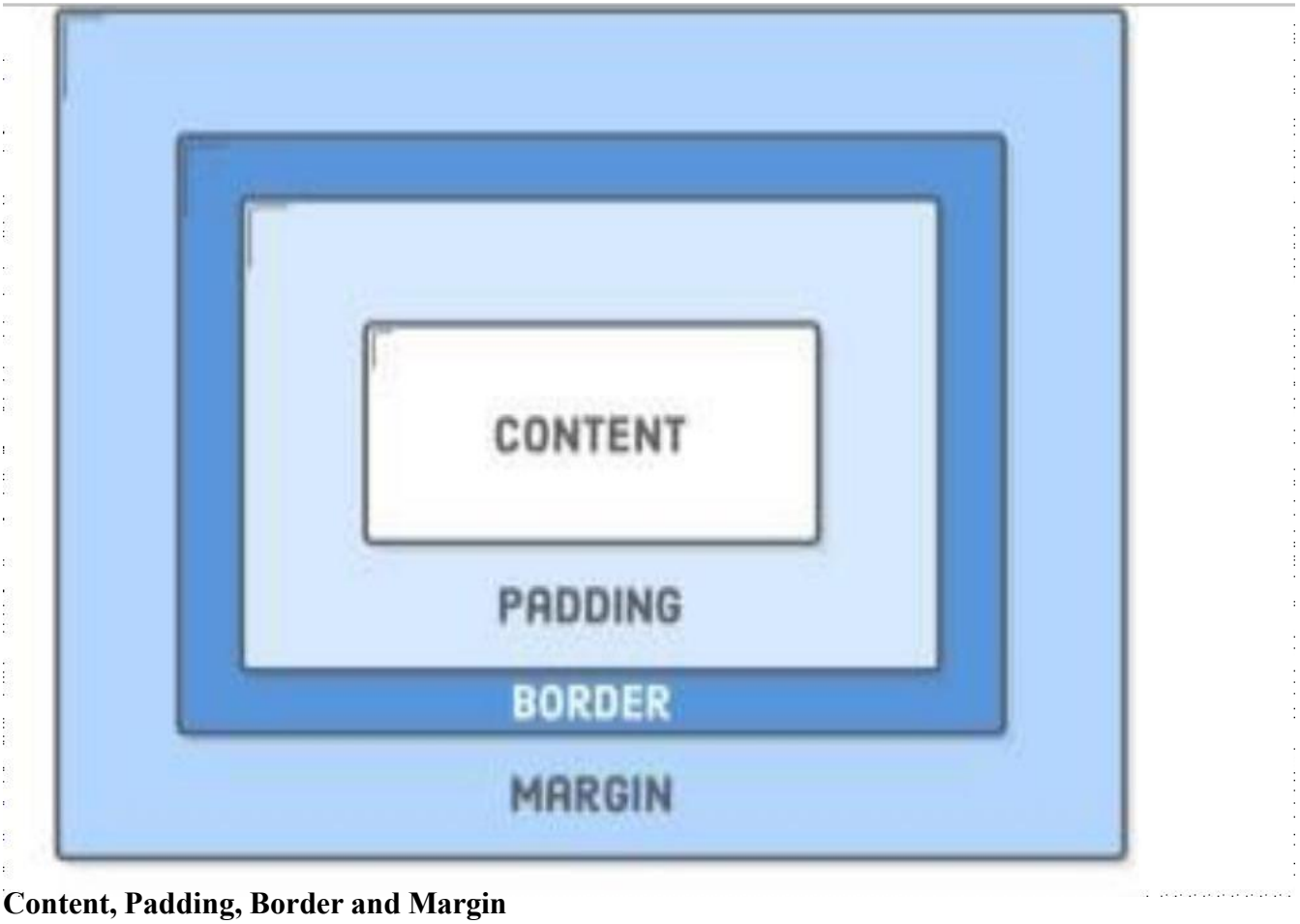
Upload Resume:
<input type="file" name="resume" accept=".pdf,.doc,.docx"><br><br>

   Upload Multiple Files:
   <input type="file" name="files" multiple><br><br>
   <input type="submit" value="Upload">
</form>
</body>
</html>
```

8. Explain the different components of the CSS Box Model with a neat diagram and an example.

The **CSS Box Model** is a layout model used in CSS to define the space occupied by an HTML element. Every HTML element is treated as a rectangular box consisting of four parts:

1. **Content**
2. **Padding**
3. **Border**
4. **Margin**



Content, Padding, Border and Margin

- **Content** – The text, image, or other media content in the element.
- **Padding** – The space between the box‘s content and its border.
- **Border** – The line between the box‘s padding and margin.
- **Margin** – The space between the box and surrounding boxes.
- Together, this is everything a browser needs to render an element‘s box.

9. Explain text input controls in HTML and discuss the various types with suitable examples.

Answer:

Text input controls are form elements used to collect user input in the form of text. There are three main types

- **Single-line text input**

Created using <input type="text">used for names, search boxes, etc.

Example: <input type="text" name="name">

- **Password input**

Created using <input type="password">hides typed characters for security.

Example: <input type="password" name="password">

- **Multi-line text input**

Created using <textarea> used for longer text like comments or messages.

Example: <textarea rows="4" cols="30"></textarea>

10. **Explain the GET and POST methods used in HTML forms and compare them with suitable examples.**

GET Method: Sends form data as part of the URL. It is mainly used to retrieve data and is not suitable for sending sensitive information.
POST Method: Sends form data in the request body. It is more secure and is used for submitting sensitive information such as passwords.

Difference between GET and POST

GET Method	POST Method
Data is sent in the URL.	Data is sent in the request body.
Less secure.	More secure.
Used to retrieve data.	Used to submit data.
Limited amount of data can be sent.	Large amount of data can be sent.

Example of GET Method:

```
<form action="login.php" method="get">
  Name: <input type="text" name="username">
  <input type="submit" value="Submit">
</form>
```

Example of POST Method

```
<form action="login.php" method="post">
  Name: <input type="text" name="username">
  <input type="password" name="password">
  <input type="submit" value="Submit">
</form>
```

MODULE - 3

Multiple Choice Questions(1 Marks): -

1. **Which of the following explains the main purpose of JavaScript in a webpage?**
   A. To Style HTML Pages
   B. To add interactivity to HTML Pages.
   C. To Perform Server Side Scripting Operation
   D. To Execute Query Related to DB on Server

2. **Which of the following describes the purpose of the <noscript> tag in HTML?**
   A. To display text for browsers that do not support JavaScript
   B. To stop JavaScript from running in supported browsers
   C. To describe non-technical movies
   D. None of the above

3. **Which of the following is a correct way to include JavaScript in a web page?**
   A. Writing code inside a <script> tag in HTML
   B. Writing code only in the browser's URL bar
   C. Writing code directly in an image file
   D. Writing code in a Word document

4. **Which of the following tasks cannot be accomplished with client-side JavaScript?**
   A. Validating a form
   B. Sending a form's contents by email
   C. Storing the form's contents to a database file on the server
   D. None of the above

5. **What can JavaScript functions do?**
   A. Only return a value
   B. Only accept parameters
   C. Accept parameters, perform operations, and return a value
   D. None of the above

6. Which of the following best illustrates the use of an array in JavaScript?
   A) let age = 20;
   B) let name = "Ravi";
   C) let marks = [70, 80, 90];
   D) let passed = true;

7. What is meant by a variable in JavaScript?
   A) A block used to repeat code
   B) A named storage location for a value
   C) A user action
   D) A collection of HTML elements

8. Which of the following best illustrates an event in JavaScript?
   A) let x = 10;
   B) 10 + 20
   C) Clicking a button
   D) let a = [1,2,3];

9. What is the main idea of using arrays in JavaScript?

A) To store multiple values in a single structure
B) To execute conditional statements
C) To handle webpage events
D) To define functions

10 . What is meant by an event in JavaScript?

A) An action that occurs
B) A type of variable
C) A type of array
D) A mathematical operator

11. How would you change the text of an HTML element with the id msg using JavaScript?

a) document.getElement("msg").innerHTML = "Text Changed!";
b) document.getElementById("msg").innerHTML = "Text Changed!";
c) document.getElementById("msg").value = "Text Changed!";
d) document.getElement("msg").text = "Text Changed!";

12. How would you write a JavaScript function to return the square of a number?

a) function square(x) { return x + x; }
b) function square(x) { return x * x; }
c) function square(x) { return x - x; }
d) function square(x) { return x / x; }

13. What is the output of the following JavaScript code?

```
var fruits = ["Apple", "Mango", "Banana"];
document.write(fruits[0]);
```

a) Apple
b) Mango
c) Banana
d) 0

14. How would you create a button to display "Hello World" in an alert box?

a) <button alert="Hello World">Click Me</button>
b) <button onclick="alert('Hello World')">Click Me</button>
c) <button onalert="Hello World">Click Me</button>
d) <button onclick="print('Hello World')">Click Me</button>

15. Write a JavaScript statement to display the current date. Select the correct statement.

a) document.date(new Date());
b) display(new Date());
c) document.write(new Date());
d) date.write(new Date());

16. Write a JavaScript statement to display "Hello World" on a webpage.

a) alert("Hello World");
b) document.write("Hello World");
c) display("Hello World");
d) message("Hello World");

17. What is the output of the following JavaScript code?

```
var fruits = ["Apple", "Mango", "Banana"];
document.write(fruits[1]);
```

a) Apple
b) Mango
c) Banana
d) 1

18. How would you write a JavaScript function to return the sum of two numbers?

a) function add(a,b) { return a-b; }
b) function add(a,b) { return a+b; }
c) function add(a,b) { return a*b; }
d) function add(a,b) { return a/b; }

19. How would you display "Welcome" in an alert box using JavaScript?

a) alert("Welcome");
b) document.write("Welcome");
c) display("Welcome");
d) message("Welcome");

20. How would you write a JavaScript function to check whether a number is even?

a) function even(n) { return n % 2 == 0; }
b) function even(n) { return n % 2 == 1; }
c) function even(n) { return n / 2 == 0; }
d) function even(n) { return n * 2 == 0; }

Short Answer Questions(3 Marks): -

1. How would you use a switch statement in JavaScript to display the current day of the week? Explain with an example program.

Answer:

The switch statement in JavaScript is used to select one block of code from multiple options based on a given value.

Syntax:

```
switch(expression) {
  case value1: // code block
    break;
  case value2: // code block
    break;
  default:      // code if no case matches
}
```

Example:

```
<script>
var day = 2;

switch(day) {
   case 1: document.write("Monday"); break;
   case 2: document.write("Tuesday"); break;
   default: document.write("Other Day");
}
</script>
```

The switch statement displays the day name based on the value of day.

2. Demonstrate how you would use a while loop in JavaScript to print numbers from 0 to 4. Explain the logic with an example.

Answer:

The while loop in JavaScript repeats a block of code as long as the condition is true**.**

Syntax:

```
while (condition) {
  // code
}
```

Example:

```
<scrip
t> let
i=0;
while(i<5){


  document.write("The number is " + i + "<br>");
  i++;
}
</script>
```

3. How would you use arrays in JavaScript to store and display the days of the week? Explain with a suitable example.

Answer:

An **array** can hold multiple values under a single variable name. Each value is accessed using an index number (starting from 0).

Ways to Declare:

```
var arr = new Array("Sun","Mon","Tue");  // Using
constructor var days = ["Sun","Mon","Tue","Wed"];        //
Using []
```

Accessing Elements:

```
document.write(days[0]); //
Sun document.write(days[3]);
// Wed
```

Example:

```
var week =
["Sun","Mon","Tue","Wed","Thu","Fri","Sat"];
document.write("First: " + week[0]);     // Sun
document.write("<br>Last: " + week[6]); // Sat
```

4. How would you get input from a user using a prompt box in JavaScript? Write a short program that asks for the user's name and displays a greeting.

Answer:

A **prompt box** is used when you want the user to **enter a value before proceeding**.

When a prompt box pops up, the user must click either **"OK"** or **"Cancel"**:

- If the user clicks **"OK"**, the box returns the **input value**.
- If the user clicks **"Cancel"**, the box returns **null**.

Syntax:

```
prompt("message", "defaultValue");
```

Example:

```
<script>
var name = prompt("Enter your name");
if(name)
{
  document.write("Hello " + name + "! How are you today?");
}
</script>
```

Output:

If you type **Tom** → Hello Tom! How are you today?

5. **How would you use the onmouseover and onmouseout events in JavaScript to change the colour of text? Provide a working example.**

Answer:

- **onmouseover**: Executes a function when the mouse pointer moves **over** an HTML element.
- **onmouseout**: Executes a function when the mouse pointer moves **out** of an HTML element.

Example:

```
<p onmouseover="this.style.color='red'" onmouseout="this.style.color='black'">
   Move the mouse over this text</p>
```

Explanation:

- onmouseover changes the text colour to red.
- onmouseout changes the text colour back to black.

6. **Demonstrate the use of variables in JavaScript by writing a program that adds two numbers and displays the result.**

Answer:

JavaScript variables are used to store data values. We can use variables to perform calculations and display results.

Example:

```
<script>
var x = 5;
var y = 6;
var sum = x + y;
document.write("Sum = " + sum);
</script>
```

7. **How would you apply a for loop in JavaScript to print numbers from 0 to 4? Explain the working with a suitable example.**

Answer:

A for loop is used to repeat a block of code a fixed number of times.

Example:

```
<script>
for (var i = 0; i < 5; i++) {
   document.write(i + "<br>");
}
</script>
```

Output:

0
1
2
3
4

Explanation: The loop starts from 0 and runs while i < 5. After each iteration, i increases by 1.

8. Write a JavaScript program to demonstrate the use of an Alert Box.

Answer:

```
<html>
<head>
<script
  type="text/javascript">
  function myFunction()
{
  alert("I am an alert box!");
  }
</script>
</head>
<body>
  <input type="button" onclick="myFunction()" value="Show alert box">
</body>
</html>
```

9. Write a JavaScript program to demonstrate the use of a Confirmation Box.

Answer:

```
<html>
<head>
   <title>Confirm Box Example</title>
</head>
<body>

<script>
   var result = confirm("Do you want to continue?");

   if (result == true) {
      document.write("You clicked OK.");
   } else {
      document.write("You clicked Cancel.");
   }
</script>
</body>
</html>
```

10. Write a JavaScript program to demonstrate the use of the onkeydown event.

Answer:

```
<html>
<body>
<input type="text" onkeydown="showMessage()">
<script>
```

```
function showMessage()
{
   alert("Key Pressed");
}
</script></body></html>
```

Long Answer Questions(5 Marks): -

1. Explain the purpose of JavaScript and describe what it can do on a web page.

Answer:

JavaScript is a scripting language used to make web pages **interactive and dynamic**. It is usually embedded in HTML pages and runs in the web browser.

Purpose of JavaScript:

- It adds **interactivity** to web pages.
- It can respond to **user actions** such as clicks and key presses.
- It can **change HTML content and elements** dynamically.
- It can **validate form data** before submission.
- It can display **alert, confirmation, and prompt boxes**.
- It can create **animations and interactive effects**.
- It can store and retrieve information using **cookies**.

Thus, JavaScript helps create **dynamic, interactive, and user-friendly web pages**.

2. Explain the different ways JavaScript makes decisions using conditional statements.

Answer:

JavaScript uses **conditional statements** to make decisions based on whether a condition is true or false.

The main types are:

- **if statement** – Executes code when a condition is true.
- **if...else statement** – Executes one block if the condition is true and another block if it is false.
- **if...else if...else statement** – Checks multiple conditions and executes the appropriate block.

Example:

```
var marks = 75;

if (marks >= 50) {
   document.write("Pass");
} else {
   document.write("Fail");
  }
```

Here, JavaScript checks the condition and displays **"Pass"** if the marks are 50 or above; otherwise, it displays **"Fail"**.

3. Explain the different locations where JavaScript code can be placed in an HTML page.

Answer:

1. Inside <head> Tag

JavaScript can be written inside the <script> tag in the <head> section.

```
<head>
<script>
function popup() {
   alert("Hello World");
}
```

```
</script></head>
```

2. Inside <body> Tag

JavaScript can be written inside the <script> tag in the <body> section.

```
<body>
<script>
function popup() {
   alert("Hello World");
}
</script></body>
```

3. External JavaScript File

JavaScript can be written in a separate **.js file** and linked to the HTML page.

myjs.js

```
function popup() {
   alert("Hello World");
}
```

HTML:

```
<script src="myjs.js"></script>
```

4. **What is a function in JavaScript? Explain how to define and call a function with an example.**

Answer:

A **function in JavaScript** is a block of code designed to perform a specific task. Functions help to **reuse code** without writing the same code repeatedly.

Syntax:

```
function functionName(parameters) {
  // code to be executed
 }
```

Example: Function Without Arguments

```
function greetings()
 {
   alert("Good Morning");
}
greetings();
```

Output: Good Morning

Example: Function With Arguments

```
function findSum(a, b)
{
   alert("Sum is: " + (a + b));
}findSum(5, 7);
```

Output: Sum is: 12

In short: A function is **defined using the function keyword** and is **called using its function name followed by parentheses ()**.

5. **Illustrate the working of Alert Box and Confirmation Box with suitable examples.**

Answer:

a) Alert Box:

An **alert box** is used to display important information to the user. It interrupts the user's activity until they acknowledge the message by clicking **OK**.

Syntax

```
alert("sometext");
```

Example

```
<html>
<head>
<script type="text/javascript">
function myFunction()
{ alert("I am an alert box!");
}
</script>
</head>
<body>
<input type="button" onclick="myFunction()" value="Show alert box" />
</body>
</html>
```

b) Confirm Box:

A confirm box asks the user to click **OK or Cancel**; it returns **true** for OK and **false** for Cancel.

Syntax

```
confirm("sometext");
```

Example

```
<html>
<head>
  <title>Confirm Box Example</title>
</head>

<body>
<script>
  var result = confirm("Do you want to continue?");

  if (result == true) {
    document.write("You clicked OK.");
  } else {
    document.write("You clicked Cancel.");
  }
</script>

</body>
</html>
```

6. Explain how the switch statement in JavaScript can be used to display the current day of the week with a suitable example program.

Answer:

The **switch statement** in JavaScript is used to execute one block of code from multiple options, based on the value of an expression.

It compares the expression with each case, and executes the matching block. If no case matches, the default block (if provided) is executed.

Syntax:

```
switch(expression) {
  case value1: // code block
    break;
  case value2: // code block
    break;
  default:      // code if no case matches
}
```

Example:

```
var day = new Date().getDay();
```

```
var x;

switch (day) {
  case 0: x = "Sunday"; break;
  case 1: x = "Monday"; break;
  case 2: x = "Tuesday"; break;
  case 3: x = "Wednesday";
  break;
  case 4: x = "Thursday"; break;
  case 5: x = "Friday"; break;
  case 6: x = "Saturday"; break;
  default: x = "Invalid day!";
}
```

7. Explain how a prompt box in JavaScript is used to get input from the user. Give a suitable example program.

Answer:

A **prompt box** is used when you want the user to **enter a value before proceeding**.
When a prompt box pops up, the user must click either **"OK"** or **"Cancel"**:

If the user clicks **"OK"**, the box returns the **input value**.
If the user clicks **"Cancel"**, the box returns **null**.

Syntax:

```
prompt("message", "defaultValue");
```

Example:

```
<script>
var name = prompt("Enter your name");
if(name){
  document.write("Hello " + name + "! How are you today?");
}
</script>
```

Output:

If you type **Tom** → Hello Tom! How are you today?

8. Explain the onkeydown and onkeyup events in JavaScript with a suitable example.

Answer:

JavaScript provides keyboard events to handle actions performed by the user using the keyboard.

1. onkeydown
   The onkeydown event occurs when the user presses a key on the keyboard.
2. onkeyup
   The onkeyup event occurs when the user releases a key from the keyboard.

Example:

```
<html>
<body>
<input type="text"
    onkeydown="showDown()" onkeyup="showUp()">

<script>
function showDown() {
   alert("Key Pressed");
}
```

```
function showUp() {
   alert("Key Released");
}
</script>
</body>
</html>
```

9. Explain the onfocus and onblur events in JavaScript with a suitable example.
Answer:

JavaScript provides **onfocus** and **onblur** events to handle the focus of an HTML element.

- **onfocus** – Occurs when an element **gains focus**, such as when the user clicks inside a text box.
- **onblur** – Occurs when an element **loses focus**, such as when the user clicks outside the text box.

Example:

```
<html>
<body>
<input type="text"
      onfocus="showFocus()"  onblur="showBlur()">

<script>

function showFocus() {
   alert("Text box is focused");
}

function showBlur() {
   alert("Text box lost focus");
}
</script>
</body>
</html>
```

10. Explain the common DOM methods used in JavaScript and describe the purpose of each method.
Answer:

The **DOM (Document Object Model)** allows JavaScript to access and change HTML elements. Some common DOM methods are:

- **getElementById()** – Finds an HTML element using its ID.
  document.getElementById("demo");

- **getElementsByTagName()** – Finds HTML elements using their tag name.
  document.getElementsByTagName("p");

- **getElementsByClassName()** – Finds HTML elements using their class name.
  document.getElementsByClassName("text");

- **querySelector()** – Finds the first element that matches a CSS selector.
  document.querySelector(".text");

MODULE – IV

Multiple Choice Questions(1 Marks) :-

1. **How would you use the Array() function to store the subjects "HTML", "CSS", and"VBScript in an array?**

A) subjects=List("HTML","CSS","VBScript")
B) subjects=Array("HTML","CSS","VBScript")
C) subjects=CreateArray("HTML","CSS","VBScript")
D) subjects=NewArray("HTML","CSS","VBScript")

2. **In the following VBScript code:**
Dim fruits
fruits = Array("Apple", "Banana", "Cherry")
You want to resize the array to add more elements without losing the existing values. What is the correct approach to preserve the array values while resizing it?

A) Dim fruits(5)
B) ReDim fruits(5)
C) ReDim Preserve fruits(5)
D) fruits = Array()

3. **In the following VBScript code:**
Dim fruits
fruits = Array("Apple", "Banana", "Cherry", "Mango")
What is the correct syntax to fetch the last element "Mango" from the array using the UBound() function?

A) fruits(UBound(fruits))
B) fruits(LBound(fruits))
C) fruits(3)
D) UBound(fruits("Mango"))

4. **In the following VBScript code:**
Dim fruits
fruits = Array("Apple", "Banana", "Cherry", "Mango")
What is the correct syntax to combine all the elements of this array into a single string separated by commas?

A) Join(fruits, ", ")
B) Split(fruits, ", ")
C) Filter(fruits, ", ")
D) Concat(fruits, ", ")

5. **How would you solve a program that contains a loop executing indefinitely?**

A) Use Do...While only
B) Insert an Exit Do condition
C) Avoid using counters
D) Remove the loop

6. **In the following VBScript code:**
   Dim fruits
   fruits = Array("Apple", "Banana", "Cherry", "Mango")
   What is the correct function to check whether the variable fruits is an array?
   A) CheckArray(fruits)
   B) IsArray(fruits)
   C) ArrayCheck(fruits)
   D) ValidateArray(fruits)

7. **Write a VBScript statement to create an array containing 1, 2, 3, 4, 5.**
   A) array1 = List(1,2,3,4,5)
   B) array1 = Array(1,2,3,4,5)
   C) array1 = CreateArray(1,2,3,4,5)
   D) array1 = NewArray(1,2,3,4,5)

8. **Write a VBScript statement used to store the value 85 at the third position of an array?**
   A) marks(2)=85
   B) marks(3)=85
   C) marks[2]=85
   D) marks=85(2)

9. **Write a VBScript condition to check whether marks is greater than or equal to 40.**
   A) If marks<=40 Then
   B) If marks<>40 Then
   C) If marks>=40 Then
   D) If marks>40 Then

10. **Write a VBScript statement to repeat a task from 1 to 10 using a For loop.**
    A) For i=1 To 10
    B) Loop i=1 To 10
    C) Repeat i=1 To 10
    D) For i=1 Until 10

11. **How would you resize an existing array named students?**
    A) Resize students
    B) ChangeDim students
    C) ReDim students(10)
    D) NewDim students(10)

12. **Which code extracts array values containing "ap"?**
    A) Filter(Array("apple","banana","apricot"),"ap")
    B) Split(Array("apple","banana","apricot"),"ap")
    C) Join(Array("apple","banana","apricot"),"ap")
    D) Find(Array("apple","banana","apricot"),"ap")

13. **How would you use the Join() function to create a comma-separated string from an array?**
    A) MsgBox Split(Array("red", "green", "blue"), ", ")
    B) MsgBox Join(Array("red", "green", "blue"), ", ")
    C) MsgBox Filter(Array("red", "green", "blue"), ", ")
    D) MsgBox LBound(Array("red", "green", "blue"))

14. How would you design a Do Until loop to display numbers from 1 to 5?

A) Do Until i > 5
B) Do Until i < 5
C) Do Until i = 5
D) Do Until i = 0

15. How would you write a VBScript loop to execute statements for each element in an array?

A) For Each item In fruits
B) For All item In fruits
C) For item With fruits
D) For item Of fruits

16. **Write a VBScript statement to combine the elements of an array named subjects using a comma.**

A) result=Split(subjects,",")
B) result=Join(subjects,",")
C) result=Combine(subjects,",")
D) result=Array(subjects,",")

17. How would you use the Split() function to convert the string "HTML,CSS,VBScript" into an array?

A)result=Join("HTML,CSS,VBScript",",")
B)result=Split("HTML,CSS,VBScript",",")
C)result=Split("HTML,CSS",",")
D) result = Split("HTML,CSS,VBScript")

18. How would you declare a variable named studentName in VBScript?

A) Var studentName
B) Dim studentName
C) Declare studentName
D) Variable studentName

19. How would you use the LBound() function to display the first index of an array named marks?

A) MsgBox UBound(marks)
B) MsgBox LBound(marks)
C) MsgBox marks(LBound)
D) MsgBox First(marks)

20. How would you display the message "Welcome" using VBScript?

A) Display "Welcome"
B) Print "Welcome"
C) MsgBox "Welcome"
D) Show "Welcome"

Short Answer Questions(3 Marks):-

1. **Write a program using a For Each loop to display all the city names stored in an array.**

Answer:
A For Each loop is ideal for iterating through array elements without needing index values.

```
Dim cities, city
cities = Array("Delhi", "Mumbai", "Chennai", "Kolkata")
For Each city In cities
MsgBox city
  Next
```

2. **Write a VBScript program to declare an array of 5 integers and calculate the sum of all elements.**

Answer:

```
Dim arr(4), i, sum
arr(0)=10
arr(1)=20
arr(2)=30
arr(3)=40
arr(4)=50
sum = 0
For i = 0 To 4
sum = sum + arr(i)
Next
MsgBox "The sum of all elements is " & sum
```

Output:
The sum of all elements is 150

3. **Demonstrate how to use a two-dimensional array to store subject names and marks.**

Answer:

```
Dim marks(1,1)

marks(0,0) = "Math": marks(0,1) = 85
marks(1,0) = "Science": marks(1,1) = 90
MsgBox marks(0,0) & " - " & marks(0,1)
```

2D arrays are helpful when storing structured/tabular data.

4. **Write a VBScript program using a loop to display all even numbers from 1 to 20.**

Answer:

```
Dim i

For i = 1 To 20

If i Mod 2 = 0 Then
MsgBox i
```

```
 End If
Next
```
Output:
Displays: 2, 4, 6, 8, …, 20 (all even numbers from 1 to 20).

5. **How would you solve the problem of adding new values to a declared array without losing existing values? Explain with an example.**

Answer:

Use ReDim Preserve to resize arrays dynamically.

```
Dim items()
ReDim items(1)
items(0) = "Pen"
items(1) = "Book"
ReDim Preserve items(2)
items(2) = "Pencil"
```

This allows you to add elements while preserving earlier data.

6. **Demonstrate how to calculate the area of a circle using constant in VBScript**.

Answer:

```
Const PI = 3.14
Dim radius, area
radius = 7
area = PI * radius * radius
MsgBox "Area: " & area
```

Using a constant like PI keeps the value fixed. The program calculates the area of a circle using πr^2.

7. **Write a VBScript program using a For loop to display a countdown from 10 to 1.**

Answer:

In **VBScript**, you can use a For loop to display a countdown from **10 to 1**:

```
Dim i
For i = 10 To 1 Step -1
MsgBox i
Next
```

8. **How would you organize subject names and marks in VBScript using two-dimensional array? Illustrate with an example.**

Answer:

A two-dimensional array stores data in rows and columns.

```
Dim marks(1,1)
marks(0,0) = "Math"
marks(0,1) = 85
marks(1,0) = "Science"
marks(1,1) = 90

MsgBox marks(0,0) & " - " & marks(0,1)
```

Output: Math - 85

This array stores subject names and their marks in rows and columns.

9. **Write a VBScript program to create an array of five numbers and display the smallest and largest index of the array using LBound() and UBound() functions.**

Answer:

```
Dim numbers
numbers = Array(10, 20, 30, 40, 50)
Document.Write "Smallest Index: " & LBound(numbers)
Document.Write "Largest Index: " & UBound(numbers)
```

LBound() returns the smallest index, while UBound() returns the largest index of the array.

10. **Write a VBScript program to create an array of three fruits and display each fruit using a For Each loop.**

Answer:

```
Dim fruits, fruit
fruits = Array("Apple", "Banana", "Mango")

For Each fruit In fruits
   Document.Write fruit & "<br>"
Next
```

Output:

Apple
Banana
Mango

Long Answer Questions(5 Marks):-

1. **How would you use a For loop in VBScript to print all odd numbers from 1 to 20?**

Answer:

```
Dim i
For i = 1 To 20
If i Mod 2 <> 0 Then
MsgBox i
End If
Next
```

This VBScript uses a For loop to iterate through numbers from 1 to 20.
The condition i Mod 2 <> 0 checks whether a number is not divisible by 2, which means the number is odd. Therefore, only odd numbers from 1 to 20 will be printed.

2. **Write a VBScript program using the Select Case statement to display the grade of the student based on the marks entered by the user.**

Answer:

```
Dim marks, grade
marks = InputBox("Enter Marks:")
Select Case marks
Case 90 To 100
grade = "A"
Case 75 To 89
grade = "B"
Case 50 To 74
grade = "C"
```

```
Case 35 To 49
grade = "D"
Case Else
grade = "Fail"
End Select

MsgBox "Your Grade is: " & grade
```

Output:
Displays the grade based on the marks entered by the user.

3. How would you use a Do While loop in VBScript to calculate the factorial of a number?

Answer:

```
Dim n, fact
n = 5
fact = 1
Do While n > 0
  fact = fact * n
  n = n - 1
Loop
MsgBox "Factorial is: " & fact
```

This program calculates the factorial of a number. A Do While loop repeatedly multiplies the value until n reaches 1. This approach is useful in mathematical problems where repeated multiplication is required.

4. Write a VBScript program to demonstrate the use of LBound, UBound, and IsArray functions.

```
Dim numbers, firstIndex, lastIndex, result
numbers = Array(10, 20, 30, 40, 50)
firstIndex = LBound(numbers)
lastIndex = UBound(numbers)
result = IsArray(numbers)
MsgBox "First index of the array: " & firstIndex
MsgBox "Last index of the array: " & lastIndex
MsgBox "Is numbers an array? " & result
```

Output:

First index of the array: 0
Last index of the array: 4
Is numbers an array? True

5. Write a VBScript program to assign grades to a student using If…ElseIf…Else statements.

Answer:

```
Dim marks

 marks = 78

If marks >= 90 Then

MsgBox "Grade A" ElseIf
 marks >= 75 Then
 MsgBox "Grade B"
 ElseIf marks >= 60 Then
 MsgBox "Grade C"
Else
MsgBox "Fail"

 End If
```

6. Write a VBScript program that uses a dynamic array to store user input values during runtime and then displays all the entered values.

Answer:

Dynamic arrays are ideal when the number of inputs is unknown at the beginning. You can use ReDim to define the array and ReDim Preserve to expand it while keeping existing values.

```
Dim items(), i
ReDim items(0)
For i = 0 To 2
 ReDim Preserve items(i)
 items(i) = InputBox("Enter item " & i)
Next
```

This script collects input and stores it in a dynamic array that grows with each iteration.

7. How would you use the Split() and Join() functions in VBScript to convert a string into an array and then combine it back into a string?

Answer:

The Split() function divides a string into an array, while Join() combines array elements into a single string.

```
Dim subjects, result
subjects = Split("HTML,CSS,VBScript", ",")
result = Join(subjects, " - ")
MsgBox result
```

Output:

HTML - CSS – VBScript

8. How would you write a VBScript program using If...ElseIf...Else statements to assign grades based on students marks?

Answer:

```
Dim marks
marks = 78
```

```
If marks >= 90 Then
   MsgBox "Grade A"
ElseIf marks >= 75 Then
   MsgBox "Grade B"
ElseIf marks >= 60 Then
   MsgBox "Grade C"
Else
   MsgBox "Fail"
End If
```

9. How would you use a Do While loop in VBScript to calculate the factorial of a number entered by the user?

Answer:

```
Dim n, fact
n = 5
fact = 1
Do While n > 0
   fact = fact * n
   n = n - 1
Loop

MsgBox "Factorial is: " & fact
```

10. How would you use arrays and a For Each loop in VBScript to display the names of five programming languages?

Answer:

A For Each loop is used to access each element of an array without using index values. It simplifies array traversal.

```
Dim languages, lang
languages = Array("HTML", "CSS", "VBScript", "JavaScript", "Python")
For Each lang In languages
   MsgBox lang
Next
```

Output:

HTML
CSS
VBScript
JavaScript

MODULE-5

MCQ Type of Questions(1 Marks)

1. **What approach would you use to structure a standard navigation bar using Bootstrap classes?**
   A. <nav class="navigationbar navbar">
   B. <nav class="navbar navbar-default">
   C. <nav class="nav navbar">
   D. <nav class="navbar default">

2. **How would you use Bootstrap to add a dropdown menu to a navigation bar ?**
   A. .dropdown
   B. .select
   C. .select-list
   D. None of the above

3. **What approach would you use to organize a group of list items with uniform styling?**
   A. .grouped-list
   B. .select-list
   C. .list-group
   D. .list-grouped

4. **How would you display an image in a circular shape using Bootstrap?**
   A. .img-rounded
   B. .img-circle
   C. .img-rounded
   D. None of the above

5. **How would you apply Bootstrap styling to display alternate table rows differently?**
   A. .tab-striped
   B. .zebra-strip
   C. .table-strip
   D. **.table-striped**
   Answer: .table-striped

6. **How would you make a button appear large using Bootstrap?**
   A. .btn-xl
   B. .btn-lrg
   C. .btn-large
   D. .btn-lg

7. **How would you open a Bootstrap modal with the ID myModal?**
   A) data-toggle="popup" data-target="#myModal"
   B) data-toggle="modal" data-target="#myModal"
   C) data-modal="open" target="myModal"
   D) modal-target="#myModal"

8. **What approach would you use to arrange three action buttons (Save, Edit, Delete) on a single horizontal line?**
   A).btn-toolbar
   B).btn-group
   C).btn-inline
   D) .btn-stack

9. **How would you add a caption to a Bootstrap carousel slide?**
   A) Use
   **B) Use **
   C) Use
   D) Use <caption class="carousel">

10. **How would you connect a Bootstrap button to a collapsible element with the ID #demo?**
    A)data-target="#demo"
    B) data-id="#demo"
    C)target-data="#demo"
    D) collapse-target="demo"

11. **What is the output of the following Bootstrap code?**
    5

    A. Displays a tooltip with 5
    B. Displays a badge showing 5
    C. Displays an alert showing 5
    D. Displays a button showing 5

12. **How would you set the progress of a Bootstrap progress bar to 70%?**
    A. height:70%
    B. size:70%
    C. width:70%
    D. progress:70%

13. **How would you write the Bootstrap code to create a vertically stacked button group?**

    A) ...

    B) ...

    **C) ... **

    D) ...

14. **How would you make an image responsive using Bootstrap?**

    A) <img class="responsive" src="photo.jpg">

    B) <img class="img-fluid" src="photo.jpg">

    C) <img class="img-responsive-table" src="photo.jpg">

    D) <img class="image-fluid-responsive" src="photo.jpg">

15. **How would you write the Bootstrap code to create a large modal?**

    A) ...

    D) ...

**C) ... **

D) ...

16. Which HTML attribute would you apply to a button to display additional information when the user hovers over it?

A) <button alt="Your tooltip text">Button</button>
B) <button title="Your tooltip text">Button</button>
C) <button href="Your tooltip text">Button</button>
D) <button value="Your tooltip text">Button</button>

17. How would you create a two-column layout using the Bootstrap grid on medium screens?
A).col-md-3
B).col-md-4
C).col-md-6
D).col-md-12

18. What is the syntax to create a Bootstrap success alert message?
A. Success!
B. Success!
**C. Success! **
D. Success!

19. **How would you create a basic Bootstrap pagination?**
a) Add the .page class to a element
b) Add the .pagination class to a element
c) Add the .paginate class to a element
d) Add the .pages class to a <nav> element

20. What is the syntax to display an unread message count using a Bootstrap badge?
A.5
B. <spanclass="alert">5
C.5
D. 5

Short Answer Questions(3 Marks)

1. Design a Bootstrap breadcrumb navigation for a university website that displays the path:Home → Departments → Computer Science → Faculty.

Answer

Bootstrap breadcrumb shows the path of a webpage.

```
<ol class="breadcrumb">
  <li class="breadcrumb-item"><a href="#">Home</a></li>
  <li class="breadcrumb-item"><a href="#">Departments</a></li>
  <li class="breadcrumb-item"><a href="#">Computer Science</a></li>
  <li class="breadcrumb-item active">Faculty</li>
</ol>
```

2. How would you apply .container and .container-fluid in Bootstrap to create different web page layouts?

Answer

- Use .container when you want a fixed-width layout with margins on both sides. Use .container-fluid when you want a full-width layout that stretches across the entire screen.
- .container → good for centered, fixed content.
- .container-fluid → good for large, responsive designs.

3. How would you use Bootstrap's grid system to design a layout that adjusts dynamically across mobile, tablet, and desktop screens?

Answer

Use Bootstrap's grid system with responsive classes like col-sm-, col-md-, and col-lg- to adjust layout for different screens.
Example:

```
<div class="row">
  <div class="col-sm-12 col-md-6 col-lg-4">Content</div>
</div>
```

Adjusts automatically for mobile (sm), tablet (md), and desktop (lg) screens.

4. Design a Bootstrap button group with three buttons — Add, Edit, and Delete — aligned horizontally, where each button has a different color style.

Answer

A button group keeps buttons together in a row.

```
<div class="btn-group">
  <button class="btn btn-success">Add</button>
  <button class="btn btn-warning">Edit</button>
  <button class="btn btn-danger">Delete</button>
</div>
```

Buttons Add, Edit, and Delete are aligned horizontally with different colors.

5. How would you use Bootstrap classes to display code snippets and lead paragraphs effectively?

Answer

Use Bootstrap's **.**lead class to make a paragraph stand out, and **<code>** to display code neatly.

```
<p class="lead">This is an important introduction paragraph.</p>
<p>Example code: <code>Hello World</code></p>
```

.lead highlights text and <code> shows code snippets clearly.

6. What approach would you use to make a table responsive and visually appealing with Bootstrap's utility classes?

Answer

Use .table for styling and .table-responsive to make it fit small screens.

```
<div class="table-responsive">
<table class="table table-striped table-bordered">
<tr>
<th>Name</th>
<th>Marks</th>
</tr>
<tr>
<td>John</td>
<td>85</td>
</tr>
</table>
</div>
```

7. Demonstrate how to change the position of a Bootstrap popover and make it close when the user clicks outside.

Anwer:

- By default, the popover appears on the right.
  Use **data-placement** to set position (top, bottom, left, right).
  Use **data-trigger="focus"** to close it when the user clicks outside.

8. Demonstrate how you would show the progress of a file upload using Bootstrap.

Answer:

- ❖ Create a .progress container.
- ❖ Add a .progress-bar inside it.
- ❖ Set the width (e.g., width:70%) to represent progress.
- ❖ Display the percentage to the user.

9. Design a toolbar for an Employee Management System that displays Add, Edit, and Delete buttons in a single horizontal row with different colors.

Answer:

```
<div class="btn-group">
<button class="btn btn-success">Add</button>
<button class="btn btn-warning">Edit</button>
<button class="btn btn-danger">Delete</button>
</div>
```

Buttons Add, Edit, and Delete are aligned horizontally with different colors.

10. What changes would you make if a developer wants a popover to appear above a button and close automatically when the user clicks outside it using Bootstrap?

Answer:

Use data-bs-placement="top" ->to display the popover above the button.

Use data-bs-trigger="focus" ->so that the popover closes automatically when the user clicks outside it.

Long Answer Questions(5 Marks)

1. **How would you customize Bootstrap cards to display dynamic content such as images, titles, and descriptions?**

Answer

Bootstrap cards can show images**,** titles**,** and descriptions using simple classes.

```
<div class="card" style="width: 18rem;">
<img src="pic.jpg" class="card-img-top" alt="...">
<div class="card-body">
<h5 class="card-title">Student Name</h5>
<p class="card-text">This is a short description.</p>
</div>
</div>
```

Explanation:

- .card makes the card box.
- .card-img-top adds an image.
- .card-title and .card-text show title and content.

2. **Demonstrate how pagination in Bootstrap helps divide long content into manageable segments.**

Answer

Bootstrap pagination helps divide long content into smaller, easy-to-view parts by adding page links.

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
</ul>
```

- .pagination creates a navigation bar for pages.
- .page-item defines each page number.
- .page-link gives clickable links.
- This helps users move between pages instead of scrolling through long content.

3. **How would you use badges to visually represent dynamic data on a dashboard?**

Answer

Bootstrap badges are used to show dynamic data like notifications, messages, or counts on a dashboard.

Eg:

```
<h4>Messages <span class="badge bg-primary">5</span></h4>
<h4>Alerts <span class="badge bg-danger">2</span></h4>
```

- .badge creates the badge shape.
- bg-primary, bg-danger, etc., give different colors.
- The number inside the badge can change dynamically using JavaScript or database data.
- Badges help quickly show live updates or counts on dashboards.

4. What approach would you use to prioritize content visibility using responsive column ordering?

Answer

Bootstrap's push and pull classes help to reorder columns and prioritize important content on different screen sizes.

Eg:

```
<div class="row">
  <div class="col-md-8 col-md-push-4">Main Content</div>
  <div class="col-md-4 col-md-pull-8">Sidebar</div>
</div>
```

.col-md-push-* moves a column forward.

.col-md-pull-* moves a column backward.

This allows important content to appear first on smaller screens, improving visibility and layout.

5. Design a Bootstrap registration form using tooltips and popovers to guide users.

Answer:

Bootstrap forms can use tooltips for short hints and popovers for detailed help while users fill in complex fields.

Eg:

```
<form>
   <input type="text" class="form-control"
        placeholder="Username"
        data-bs-toggle="tooltip"
        title="Enter your username">

   <input type="password" class="form-control mt-2"
        placeholder="Password"
        data-bs-toggle="popover"
        title="Password Help"
        data-bs-content="Use at least 8 characters">

   <button class="btn btn-primary mt-2">Register</button>
</form>
```

Tooltips show small hints like "Enter your name".

Popovers show detailed help like password rules.

6. Demonstrate how to implement pagination in Bootstrap to navigate between different pages of content.

Answer

Bootstrap pagination helps divide long content into smaller, easy-to-view parts by adding page links.

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
```

```
  <li class="page-item"><a class="page-link" href="#">3</a></li>
</ul>
```

- .pagination creates a navigation bar for pages.
- .page-item defines each page number.
- .page-link gives clickable links.
- This helps users move between pages instead of scrolling through long content.

7. How would you implement Bootstrap alerts on a webpage to display different types of important messages?

Answer

Bootstrap alerts are used to show important messages like success, warning, or error on a webpage.

Eg:

```
<div class="alert alert-success">Registration Successful!</div>
<div class="alert alert-warning">Please check your input.</div>
<div class="alert alert-danger">Error! Try again later.</div>
```

- .alert creates the alert box.
- Classes like .alert-success, .alert-warning, and .alert-danger show different message types with different colors.
- Helps users quickly understand the type and importance of the message.

8. How would you use Bootstrap alerts to provide users with different types of feedback after performing an action?

Answer

Bootstrap alerts are used to show important messages like success, warning, or error on a webpage.

```
<div class="alert alert-success">Registration Successful!</div>
<div class="alert alert-warning">Please check your input.</div>
<div class="alert alert-danger">Error! Try again later.</div>
```

- .alert creates the alert box.
- Classes like .alert-success, .alert-warning, and .alert-danger show different message types with different colors.
- Helps users quickly understand the type and importance of the message.

9. How would you create a Bootstrap card to display a student's profile with an image, name, and short option?

Answer

Bootstrap cards can show images, titles, and descriptions using simple classes.

```
<div class="card" style="width: 18rem;">
  <img src="pic.jpg" class="card-img-top" alt="...">
  <div class="card-body">
    <h5 class="card-title">Student Name</h5>
    <p class="card-text">This is a short description.</p>
  </div>
</div>
```

Explanation:

- .card makes the card box.
- .card-img-top adds an image.

- .card-title and .card-text show the title and content.

10. How would you implement Bootstrap pagination to improve navigation on a webpage containing a large amount of content?

Answer

Bootstrap pagination helps divide long content into smaller, easy-to-view parts by adding page links.

```
<ul class="pagination">
  <li class="page-item"><a class="page-link" href="#">1</a></li>
  <li class="page-item"><a class="page-link" href="#">2</a></li>
  <li class="page-item"><a class="page-link" href="#">3</a></li>
</ul>
```

- .pagination creates a navigation bar for pages.
- .page-item defines each page number.
- .page-link gives clickable links.
- This helps users move between pages instead of scrolling through long content.